Third assignment

Problem: Software engineers use issue tracking systems to discuss different design issues. Currently, we have a classification of issues regarding the types of design decisions. This is however an abstract view for the design issues and does not give a guide about the topics discussed in issues. This prevents us from creating tools that help software engineers search for certain design issues.

Goal: Explore the topics of design issues using topic modeling algorithms, and explore their characteristics and co-occurrences with the decision types.

Each group will focus on their project from assignment 1, and with a specific list of issues (attached). Each issue is classified based on three types of design decisions.

To achieve this goal, I propose these guiding steps in the following weeks:

1st week: **Download issue data and create vocabulary**.
In this week, you will prepare the data from issues and perform initial pre-processing of issue description. You will make the following steps:

1. *Use the Jira API to determine the parent of an issue*: Please note that most issues do not have parents.
2. *Download issue data using jira API:* Specifically, you need to write a python program that downloads and stores issue summary, description, all fields such as type and status, as well as number of comments. Please refer to the lecture for full data structure. Store this data in a good format as you will need it in the next steps.
3. *Perform tokenization and pre-processing for issue summary and description*:
   - Concatenate both issue summary and description.
   - Perform text cleaning and tokenization for the concatenation of issue summary and description. For this, you can use or (re-use) the Rust code in this repository. The Rust code can be called from python (e.g., this code).
   - Remove stop words.
   - Perform lemmatization and/or stemming.
4. *Create the vocabulary*: Based on the pre-processed issues, create a list of unique tokens in all issues, and count the number of occurrences of each token in all issues to create the vocabulary of the corpus.
   - What tokens have the most frequency?
   - Are there tokens candidates to be pre-processed, removed, or replaced with an ontology class?
5. *Create document term matrix for all issues after pre-processing*. This will be the input for the LDA topic modeling algorithm. See lecture.

Notes on running Rust code: To run Rust code from python, follow these steps:

1. Install Rust
2. clone the dl_manager repo
3. run `pip install nltk setuptools_rust`
4. run `python setup.py build_ext --inplace`

2nd week: **Run topic modeling algorithms on issue summary and description to determine topics.**

This week, you will run two topic modeling algorithms (LDA and BERTopic) on the list of tokens per issue.

For the LDA, you need to perform the following steps:

1. *Run LDA*: It is very important to create useful topics. This can be done, if you adjust the tokenization and pre-processing to remove or replace tokens.
   Examples of useful topics:
   - Issues on security problems.
   - Issues on technology upgrades.
   - Issues on refactoring.
   - Issues on adding components.

   Examples of non-useful topics:
   - Issues on features of Cassandra.
   - Issues that focus on specific classes or methods.

   To achieve useful topics, you will follow an iterative and experimental approach with different parameters. For example:
   - ***Iteration 1***: Run LDA with a big number of topics (e.g., 8 or 10 topics). Use low values of α and β (e.g., 0.01), because we do not expect many topics in an issue and not many terms to express a topic. Report the top terms per topic. Are the topics useful? Or do they discuss project specific features? Note: You can use the guide of this repository to write your code.

   - ***Iteration 2***: If the topics are not useful, one option is to replace terms with their respective ontology classes. These could be useful ontology classes:
     - *Component*: presents processing units on different abstractions (e.g., machine, service, class, method).
     - *Connector*: presents communications between components. They usually come as verbs (e.g., send, write, retrieve).
     - *Data*: presents data stored or transferred (e.g., message, object, dump).
     - *Solutions*: names of well-known solutions such as patterns (e.g., layer, MVC) and tactics (e.g., replication, authentication).
     - *Quality attributes*: such as performance and security. These could come in many terms. See this repo for more examples.
       One option that you create separate classes for each quality attribute. This might produce more detailed topics.

     Further examples of terms for these ontology classes are in this repo.
     Try to replace terms that have the highest frequency, because these will impact the topics. Also, only replace terms that you are sure about. For terms that do not belong to any ontology classes, you can leave them as they are.
     After replacing terms with their ontology classes, you can run the LDA again.
     Report the top terms per topic. Did the topics change?

2. *Optimize the number of topics and analyze results*, you will adjust the number of topics. First check if there are topics with few terms in them. Second, calculate and plot the **coherence score** for different numbers of topics (e.g., 3 to 10) to determine the number of topics with the highest coherence score (example of

code). Also determine **topic proportions** per issue to determine how many topics come in an issue, if topics are dominant (Example of code in this repository).

For the BERTopic, you need to perform the following steps:

1.

Based on these steps, you can answer the following research question:

Research question 1: *What topics emerge from LDA and BERTopic and what are common keywords per topic?*
This is the main result of LDA and BERTopic, which contains the top list of keywords per topic, and a simple definition about the topic. The question needs to present how many issues discuss each topic.

3rd week, **Conduct significance tests on the characteristics of topics and their co-occurrences.**
This week, you will analyze the issues of each topic and their co-occurrences to answer the following research questions:

Research question 2: *What are the characteristics of the issues in each topic?*
Here, you will analyze the fields that belong to each topic. For example, the number of comments and attachments for each topic, and the issue types for each topic. Also, the size of the description of each topic. Charts like box plots and significance tests are required to be applied.

Research question 3:
What topics significantly co-occurrence together among LDA topics and BERTopics?
What topics from LDA and BERTopics significantly co-occurre with the types of design decisions?
Here, you will apply significance tests on the co-occurrences between LDA topics, BERT topics, and the three types of design decisions.